

Bangladesh University of Engineering and Technology
Department of Electrical and Electronic Engineering
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Project Report

Project Name:

MAXWELL'S EQUATIONS SIMULATION USING 1D AND 2D FDTD

Presented by:

MD. TABASSUM TALHA

ABRESHAM NIHAL

Student's ID:

2106109

2106124

Department: EEE

Section: B2

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Course Teacher:

1. Tanvir Hossain

2. Md. Adib Rahman

Abstract

This report presents a detailed educational implementation of the finite-difference time-domain (FDTD) method for solving Maxwell's equations in one and two spatial dimensions. The one-dimensional model describes normally incident plane-wave propagation through air, dielectric slabs, and multilayer media. The two-dimensional model extends the method to spatially varying structures, scattering objects, and wave interactions in an x-y computational domain. The report develops the governing equations, Yee-grid discretization, stability criterion, source excitation, absorbing boundaries, material modeling, and frequency-domain analysis. MATLAB-oriented pseudocode is included to clarify a practical implementation path.

The central idea of FDTD is to replace derivatives in Maxwell's curl equations with finite differences and then advance electric and magnetic fields alternately in time. A Gaussian pulse is used as a broadband source. The simulated fields can be used to study propagation, reflection, transmission, diffraction, interference, and resonant behavior. The report also identifies common numerical errors and outlines extensions to lossy, dispersive, anisotropic, and three-dimensional electromagnetic models.

Introduction

Maxwell's equations provide the fundamental mathematical description of classical electromagnetic phenomena. They explain the generation and propagation of electric and magnetic fields and form the basis of many technologies, including antennas, optical devices, radar, wireless communication, microwave circuits, waveguides, sensors, and photonic structures.

Many electromagnetic problems cannot be solved analytically when the geometry contains several materials, irregular boundaries, or time-varying fields. The finite-difference time-domain method addresses this difficulty by discretizing the computational region into small cells and calculating field values at successive time steps. Because it works directly in the time domain, a single short broadband excitation can reveal the response over a wide frequency range.

This project follows a report format suitable for an EEE numerical-technique laboratory. It begins with the theoretical basis of Maxwell's equations, then develops both 1D and 2D FDTD formulations, describes an implementation workflow, presents representative test cases, explains validation methods, and discusses practical limitations.

Objectives

The purpose of this project is to build a clear numerical understanding of electromagnetic-wave propagation and to demonstrate how Maxwell's equations can be solved computationally rather than only analytically.

- To derive the 1D FDTD update equations for a plane electromagnetic wave.
- To formulate the 2D transverse-magnetic-to-z (TM_z) FDTD scheme using E_z , H_x , and H_y field components.
- To explain the Yee staggered grid and leapfrog time stepping.
- To choose stable spatial and temporal resolutions using wavelength and Courant criteria.
- To implement a Gaussian-pulse source, absorbing boundary treatment, material profiles, and DFT-based spectral analysis.

Theory

In SI units, Maxwell's microscopic equations are written as:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$
$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} = \epsilon \frac{\partial \mathbf{E}}{\partial t}$$

For a linear, isotropic medium, the constitutive relations are:

$$\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}, \quad \mathbf{J} = \sigma \mathbf{E}$$

Here, $\mathbf{E}$ is electric field intensity, $\mathbf{H}$ is magnetic field intensity, $\mathbf{D}$ is electric flux density, $\mathbf{B}$ is magnetic flux density, ρ is volume charge density, $\mathbf{J}$ is current density, ϵ is permittivity, μ is permeability, and σ is conductivity. In source-free, lossless regions, the curl equations become the principal equations used by FDTD.

For a homogeneous lossless medium, the phase velocity, intrinsic impedance, and refractive index are:

$$v = 1/\sqrt{(\mu\epsilon)}, \quad \eta = \sqrt{(\mu/\epsilon)}, \quad n = \sqrt{(\epsilon_r \mu_r)}$$

In vacuum, the speed of light is $c_0 = 1/\sqrt{(\mu_0 \epsilon_0)} \approx 2.99792458 \times 10^8$ m/s. A wave entering a dielectric generally changes wavelength and phase velocity. At an interface, impedance mismatch produces reflection and transmission.

Solution Methodologies

The following methods are used in the simulation framework:

- **FDTD:** Finite Difference Time Domain method for updating electromagnetic fields on a discrete grid.
- **Yee grid:** Spatial staggering of electric and magnetic field samples so that curl operations are represented naturally.
- **Leapfrog time stepping:** Electric and magnetic fields are evaluated at interleaved half-time steps.
- **Absorbing boundary condition:** A boundary treatment that reduces nonphysical reflections from the computational edges.
- **TF/SF excitation:** Total-field/scattered-field injection for launching an incident wave into a selected region.
- **DFT:** Discrete Fourier transform for converting recorded time-domain fields into reflection and transmission spectra.

1D FDTD Formulation

For a one-dimensional wave propagating along the z -axis, one convenient field polarization is $\mathbf{E}(z,t)$ and $\mathbf{H}_x(z,t)$. Assuming no free current and a lossless medium, Maxwell's curl equations reduce to:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \frac{\partial E_y}{\partial z}$$

$$\frac{\partial E_y}{\partial t} = \frac{1}{\epsilon} \frac{\partial H_x}{\partial z}$$

The signs depend on the selected coordinate orientation. A consistent sign convention must be maintained throughout the update equations and source injection. The essential feature is that spatial changes in E create H, and spatial changes in H create E. In the 1D Yee arrangement, E is stored at integer spatial indices and integer time indices, while H is stored halfway between adjacent E samples and at half-time steps. This staggering makes centered finite differences possible.

E	H	E	H	E	H	E	H	E	H	E	H
0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5

Figure 1. Conceptual 1D Yee grid: electric and magnetic samples are staggered in space.

Let Δz be the cell size and Δt be the time step. The lossless update equations are:

$$\frac{H_x^{k+1/2}(n+1/2) - H_x^{k-1/2}(n+1/2)}{\Delta t} = \frac{1}{\mu} \left(\frac{E_y^k(n+1) - E_y^k(n)}{\Delta z} \right)$$

$$\frac{E_y^{k+1}(n) - E_y^k(n)}{\Delta t} = \frac{1}{\epsilon} \left(\frac{H_x^{k+1/2}(n+1/2) - H_x^{k+1/2}(n-1/2)}{\Delta z} \right)$$

For a conductive medium, the E-field update includes σ . Defining coefficients C and C gives:

$$H_x(n) \leftarrow H_x(n) + \frac{m_{H_x}(n)}{\Delta z} [E_y(n+1) - E_y(n)]$$

$$E_y(n) \leftarrow E_y(n) + \frac{m_{E_y}(n)}{\Delta z} [H_x(n) - H_x(n-1)]$$

Grid and Time Resolution

FDTD accuracy depends strongly on the spatial cell size. The smallest wavelength occurs at the maximum frequency and in the material with the largest refractive index. If f is the highest frequency of interest, then:

$$\lambda = c / [f_{\max} n_{\max}]$$

A common engineering rule is to use at least 10 to 20 cells per smallest wavelength. For improved accuracy in an instructional simulation, 20 to 50 cells per wavelength may be selected:

$$\Delta z \leq \lambda / N_{\min}$$

For 1D FDTD, the Courant stability condition is:

$$\Delta t \leq \Delta z / v$$

Since the highest wave speed is normally c_0 , a safe practical choice is $\Delta t = S \Delta z / c_0$, where $0 < S \leq 1$. A choice such as $S = 0.5$ is stable and conservative.

Source Excitation

A Gaussian pulse is a useful broadband source because its Fourier transform covers a continuous frequency band. A typical source is:

$$s(t) = \exp\{-(t - t_0)^2/\tau^2\}$$

The pulse is injected at a chosen grid index. A soft source adds the signal to the field, while a hard source directly assigns the field value. A soft source is often preferred because it causes less numerical disturbance. For example:

```
source = exp(-((t - t0)/tau).^2);  
Ey(isrc) = Ey(isrc) + source(n);
```

1D Boundary Conditions

Without an absorbing boundary, a wave reaches the end of the grid and reflects back into the simulation. That reflection is numerical rather than physical. A simple first-order absorbing boundary, often called a Mur-type or perfectly absorbing boundary approximation, uses previously stored field values to approximate an outgoing wave. It is easy to implement but does not eliminate all reflection.

For higher accuracy, perfectly matched layers (PML) are preferred. A PML is an artificial lossy region designed to absorb waves at many angles and frequencies with very small reflection. In this report, a simple absorbing boundary is suitable for educational 1D simulations, while PML is recommended for serious 2D work.

1D Algorithm

The following procedure summarizes a typical one-dimensional simulation.

1. Define c_0 , ϵ_0 , μ_0 , $f_{\max}$, material parameters, and domain length.
2. Find $n_{\max} = \max(\sqrt{\epsilon_r} \cdot \mu_r)$.
3. Select Δz from the smallest wavelength criterion.
4. Select Δt from the Courant condition.
5. Create arrays E_y , H_x , ϵ_r , μ_r , and optional σ .
6. Calculate update coefficients.
7. For each time step n :
 - a. Update H_x at all interior H locations.
 - b. Apply the H-side source correction if using TF/SF.
 - c. Apply H boundary treatment.
 - d. Update E_y at all interior E locations.
 - e. Apply the E-side source correction or soft source.
 - f. Apply E boundary treatment.
 - g. Record monitor fields and update DFT accumulators.
8. Calculate reflection, transmission, and total response.

MATLAB-Oriented 1D Pseudocode

```
for n = 1:Nt      Hx(1:Nz-1) = Hx(1:Nz-1) -  
Ch(1:Nz-1) .* ...  
    (Ey(2:Nz) - Ey(1:Nz-1));  
  
    Hx(isrc-1) = Hx(isrc-1) - Ch(isrc-1) * source(n);  
  
    Ey(2:Nz-1) = Ca(2:Nz-1).*Ey(2:Nz-1) - ...  
        Cb(2:Nz-1).*(Hx(2:Nz-1) - Hx(1:Nz-2));  
  
    Ey(isrc) = Ey(isrc) + source(n);  
  
    % Apply absorbing boundaries and record monitors  
end
```

Reflection and Transmission

Field monitors are placed on the incident side and transmission side of the structure. Their recorded time-domain values are transformed at the frequencies of interest. If $S(f)$, $R(f)$, and $T(f)$ are the complex Fourier amplitudes of source, reflected, and transmitted fields, the power-like normalized quantities may be estimated as:

$$\text{Reflectance:} = |R(f)/S(f)|^2$$

$$\text{Transmittance:} = |T(f)/S(f)|^2$$

For a lossless structure with consistent normalization and sufficiently effective boundaries, the energy check should approximately satisfy:

$$R(f) + T(f) \approx 1$$

When the media on the two monitor sides have different impedances, the transmittance must include the appropriate impedance or Poynting-vector normalization rather than only a squared electric-field ratio.

2D FDTD Formulation

The two-dimensional formulation enables simulation of structures that vary in both x and y directions. Examples include apertures, dielectric blocks, metallic objects, bends, cavities, antennas, and scattering obstacles. A common 2D polarization is TM_z, in which E_z , H_x , and H_y are nonzero while all z-derivatives are zero. For source-free, lossless TM fields, Maxwell's equations reduce to:

$$\frac{\partial H_x}{\partial t} = -\frac{1}{\mu} \frac{\partial E_z}{\partial y}$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \frac{\partial E_z}{\partial x}$$

$$\frac{\partial E_z}{\partial t} = \frac{1}{\epsilon} \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right)$$

The 2D Yee cell staggers E_x , H_y , and H_z spatially. E_x is stored at the cell center, H is displaced by half a cell in y, and H is displaced by half a cell in x. y

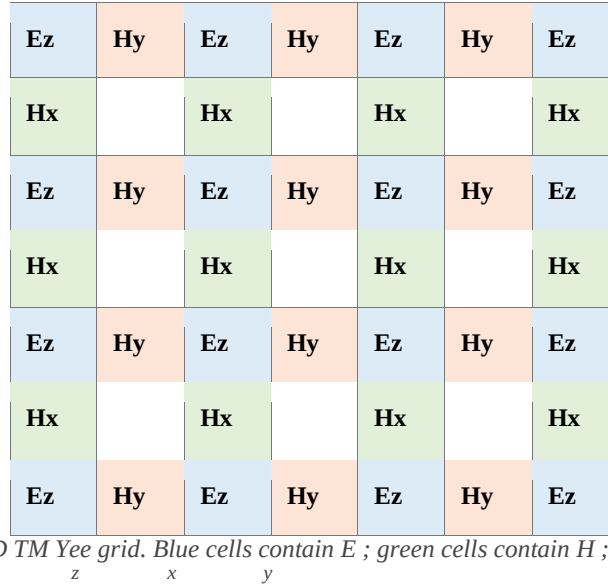


Figure 2. Conceptual 2D TM Yee grid. Blue cells contain E_z ; green cells contain H_x ; orange cells contain H_y .

For uniform grid spacings Δx and Δy , the update equations are:

$$H_x^{k+1/2}(i, j) = H_x^{k-1/2}(i, j) - \left(\frac{c_0 \Delta t}{\Delta y} \right) [E_z^k(i, j+1) - E_z^k(i, j)] \quad (20)$$

$$H_y^{k+1/2}(i, j) = H_y^{k-1/2}(i, j) + \left(\frac{c_0 \Delta t}{\Delta x} \right) [E_z^k(i+1, j) - E_z^k(i, j)] \quad (21)$$

$$E_z^{k+1}(i, j) = E_z^k(i, j) + \frac{c_0 \Delta t}{\epsilon_r(i, j)} \left(\frac{H_y^{k+1/2}(i, j) - H_y^{k+1/2}(i-1, j)}{\Delta x} - \frac{H_x^{k+1/2}(i, j) - H_x^{k+1/2}(i, j-1)}{\Delta y} \right) \quad (22)$$

2D Stability and Materials

The 2D Courant condition is more restrictive than the 1D case. In a medium whose maximum wave speed is c , stability requires:

$$\Delta t \leq 1 / [c \sqrt{(1/\Delta x^2) + (1/\Delta y^2)}]$$

When $\Delta x = \Delta y = \Delta$, a safe choice is:

$$\Delta t = S \Delta / (c \sqrt{2}), \quad 0 < S \leq 1$$

The material geometry is modeled by assigning arrays for relative permittivity $\epsilon(i, j)$, relative permeability $\mu(i, j)$, and, where required, conductivity $\sigma(i, j)$. A dielectric rectangle, circular inclusion, or multilayer pattern can be created by changing entries in these arrays. The update coefficients must then be calculated locally, cell by cell.

2D Sources

A point source can be added at one cell to produce cylindrical waves in 2D. A line source can launch a nearly planar wave. For scattering simulations, a total-field/scattered-field boundary is helpful: it separates the region containing incident plus scattered field from the outer region containing scattered field only. A simpler initial demonstration can use a soft E_z source at a single cell.

2D Boundary Conditions

Two-dimensional boundaries require more care because waves can strike at oblique angles. Simple first-order absorbing boundaries may produce noticeable reflections. A PML surrounding all four sides of the domain is the standard choice for reliable simulations. If PML is not implemented, enlarge the air buffer around the object and stop the simulation before boundary reflections return to the region of interest.

MATLAB-Oriented 2D Pseudocode

```
for n = 1:Nt      Hx(:,1:Ny-1) = Hx(:,1:Ny-1) -  
Chx(:,1:Ny-1) .* ...  
    (Ez(:,2:Ny) - Ez(:,1:Ny-1));  
  
    Hy(1:Nx-1,:) = Hy(1:Nx-1,:) + Chy(1:Nx-1,:) .* ...  
(Ez(2:Nx,:) - Ez(1:Nx-1,:));  
  
    curlH = (Hy(2:Nx,2:Ny-1) - Hy(1:Nx-1,2:Ny-1))/dx - ...  
(Hx(2:Nx-1,2:Ny) - Hx(2:Nx-1,1:Ny-1))/dy;  
  
    Ez(2:Nx-1,2:Ny-1) = Ca.*Ez(2:Nx-1,2:Ny-1) + Cb.*curlH;  
  
    Ez(isrcx,isrcy) = Ez(isrcx,isrcy) + source(n);  
  
    % Apply PML/ABC and save field snapshots  
end
```

Algorithm

A GUI-based program may allow the user to choose a medium or device, specify ϵ , μ , conductivity, r geometry, source position, simulation duration, and frequency range. The program should then generate the grid, run the solver, display field evolution, plot spectral response, and export data.

Suggested GUI Inputs

- Simulation dimension: 1D or 2D.
- Maximum frequency, grid resolution, and time-step safety factor.
- Medium type or custom values of ϵ , μ , and σ .
- 1D slab thickness or 2D object geometry and position.
- Source type: Gaussian pulse, sinusoidal source, point source, or line source.
- Boundary type: simple absorbing boundary or PML.
- Monitor locations and DFT frequency range.

Main Computational Sequence

```
Input material and geometry information
↓
Determine nmax and smallest wavelength
↓
Choose spatial grid: dz in 1D; dx and dy in 2D
↓
Choose stable time step from Courant condition
↓
Initialize field arrays and material coefficients
↓
Generate Gaussian source
↓
Repeatedly update H fields, then E fields
↓
Apply source and absorbing boundary conditions
↓
Record monitors and accumulate DFT values
↓
Calculate reflectance, transmittance, and field snapshots
↓
Plot and export results
```

Recommended Output Figures

- 1D E and H fields versus distance at selected time steps.
- 1D reflectance, transmittance, and total power response versus frequency.
- 2D color map of E (x,y) at multiple time steps. z
- 2D snapshots showing incident wave, reflection, scattering, and transmitted wave.
- Line-monitor field versus time and its DFT spectrum.

Sample Test Cases

The following test cases demonstrate how the 1D and 2D formulations can be evaluated. Numerical results should be compared with physical expectations and, where possible, analytical solutions.

1. Air – Dielectric Slab – Air (1D)

A Gaussian pulse travels through air and strikes a finite dielectric slab. At normal incidence, the approximate field reflection coefficient at a single interface is:

$$\Gamma = (\eta_2 - \eta_1) / (\eta_2 + \eta_1)$$

For nonmagnetic media, $\eta = \eta_0 / \sqrt{\epsilon_r}$. A larger dielectric constant creates stronger impedance mismatch and usually stronger reflection. A finite slab has two interfaces, so multiple internal reflections produce frequency-dependent interference in the reflectance and transmittance curves.

2. Air – Multilayer Device – Air (1D)

A three-layer structure such as paper–graphite–glass can be approximated by adjoining regions with different ϵ , μ , and, if appropriate, conductivity. The response depends on layer thickness, impedance mismatch, absorption, and phase delay. A multilayer device can show pass bands, stop bands, and resonance-like variations in its frequency response.

3. Dielectric Block Scattering (2D)

A 2D point or line source illuminates a dielectric block in air. The E field map should show the z incoming field, reflection from the block, transmission through the block, and diffraction around its edges. The field pattern changes with dielectric constant, object dimensions, source frequency content, and the amount of surrounding air buffer.

4. Circular Inclusion or Metallic Obstacle (2D)

A circular dielectric inclusion illustrates curved-interface scattering. A highly conducting or PEC-like object strongly reflects the incident field and produces a shadow region behind it. For a perfect electric conductor in TM polarization, E is forced to zero inside the conducting object. Such examples provide z intuitive demonstrations of scattering and boundary conditions.

Validation Checks

- In uniform air, the pulse should propagate at approximately c without large amplitude growth.
- The field should remain bounded; exponential growth indicates violation of the stability condition or a sign/indexing error.
- For a lossless 1D device, T + S should be close to one after proper normalization.
- Refining the grid should make the result converge rather than change unpredictably.
- When the object is removed, the simulation should reproduce free-space propagation.

Dealing with Problems

FDTD is conceptually direct, but practical simulations can be affected by numerical and implementation errors. The following issues should be addressed explicitly.

1. Incorrect maximum refractive index

Using $n = 1$ for all materials is not valid when ϵ or μ exceeds one. The grid must be designed using $\max_r$ the largest refractive index in the domain:

$$n = \max[\sqrt{(\epsilon \mu)}]_{\max}$$

If this is ignored, the cells can be too coarse inside high-index media, causing phase error and poor spectral accuracy.

2. Courant instability

The FDTD time step must obey the 1D or 2D Courant limit. If Δt is too large, fields may grow without bound. A smaller safety factor gives increased stability margin but increases run time.

3. Numerical dispersion

FDTD waves travel at a slightly grid-dependent numerical speed. When cells are too large compared with wavelength, short-wavelength components accumulate phase error. Increasing cells per wavelength reduces this numerical dispersion.

4. Boundary reflections

Simple absorbing boundaries are imperfect. Reflections from domain edges can contaminate the measured response. Use a larger buffer, stop the simulation before returns arrive, or implement PML.

5. E–H amplitude interpretation

Electric and magnetic field amplitudes have different units and are related by the local impedance, not by equality. Do not force H to equal E inside a physical simulation only for plotting convenience. Instead, normalize the displayed curves or scale H by η when visual comparison is required.

6. Frequency-response accuracy

A finite-duration simulation limits frequency resolution. Longer records improve spectral resolution, while a properly delayed Gaussian source reduces unwanted low-frequency artifacts. Reference simulations and time-windowing can improve calculation of reflection and transmission.

7. Indexing and coefficient placement

On a Yee grid, ϵ , μ , and σ must be associated with the correct field location. In 2D, careless array dimensions or staggered indexing can cause incorrect fields even if MATLAB executes without errors.

Discussion and Future Work

The 1D FDTD method is an effective first model for understanding the numerical solution of Maxwell's equations. It directly demonstrates field propagation, source injection, material interfaces, wave impedance, reflection, transmission, and spectral response. Its computational cost is small, allowing parameter sweeps and clear visualization.

The 2D extension introduces substantially richer physics. It allows spatially shaped objects and wave effects such as scattering, diffraction, interference, focusing, cavity resonance, and edge effects. It also requires greater attention to memory use, absorbing boundaries, stability, and visualization. Compared with 1D, a 2D model has many more cells and therefore many more field updates at every time step.

Several improvements can extend this work:

- Implement convolutional PML or split-field PML around all boundaries.
- Add conductivity and compare lossy-material absorption with lossless propagation.
- Model frequency-dependent dispersive media using Debye, Drude, or Lorentz material models.
- Add total-field/scattered-field boundaries for accurate plane-wave incidence in 2D.
- Use S-parameters and Poynting-vector monitors for rigorous power calculations.
- Develop a MATLAB App Designer interface for geometry editing and animated field maps.
- Extend the solver to 3D FDTD for antennas, waveguides, and full electromagnetic devices.

Conclusion

This report has developed the theoretical and practical framework for solving Maxwell's equations using both one-dimensional and two-dimensional FDTD methods. The method replaces spatial and temporal derivatives with finite differences on a staggered Yee grid. By updating magnetic fields and electric fields

alternately, the numerical model reproduces electromagnetic-wave propagation through free space and material structures.

A reliable simulation requires suitable grid resolution, a Courant-stable time step, physically consistent material properties, a well-defined source, and absorbing boundaries. The 1D method is appropriate for normal-incidence slab and multilayer problems, while the 2D TM formulation supports scattering and z geometry-dependent wave studies. With appropriate validation, FDTD is a flexible and powerful foundation for more advanced electromagnetic simulation.

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